



Zero-Knowledge Adaptive AI Agent

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Project Overview

Context & Background

Large Language Models (LLMs) have enabled advanced conversational systems capable of generating fluent and context-aware responses. These systems are increasingly being explored for deployment in advisory domains such as academia, enterprises, and policy-driven environments.

Challenges with Current LLM-based Systems



Implicit Knowledge

Depend primarily on implicit knowledge acquired during pre-training.



Grounding Gap

Produce responses without guaranteed grounding in verifiable sources.



Limited Control

Lack explicit control over the scope and boundaries of their knowledge.

Problem Statement

Academic institutions provide policy info via static docs/portals. While authoritative, they lack natural language access and context. Users must manually search for exam or grading rules, leading to errors. Generic chatbots fail due to hallucinations or lack of conversational intelligence.

Goal: Design a secure, grounded, and adaptive academic-policy assistant that can:



Authorized Knowledge

Answer only from verified institutional sources to ensure factual grounding.



Context & Preference Aware

Provide intelligent assistance without compromising factual integrity.



Multi-User Interaction

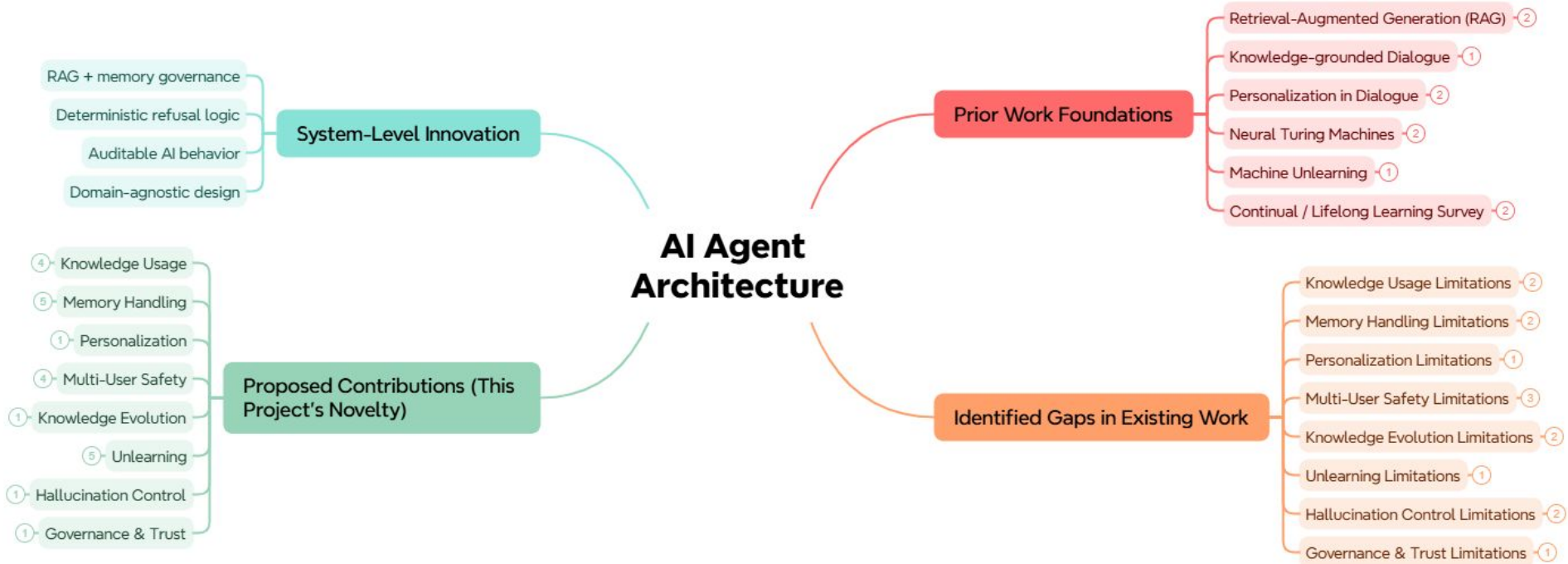
Support authenticated access and personalized interaction for diverse users.



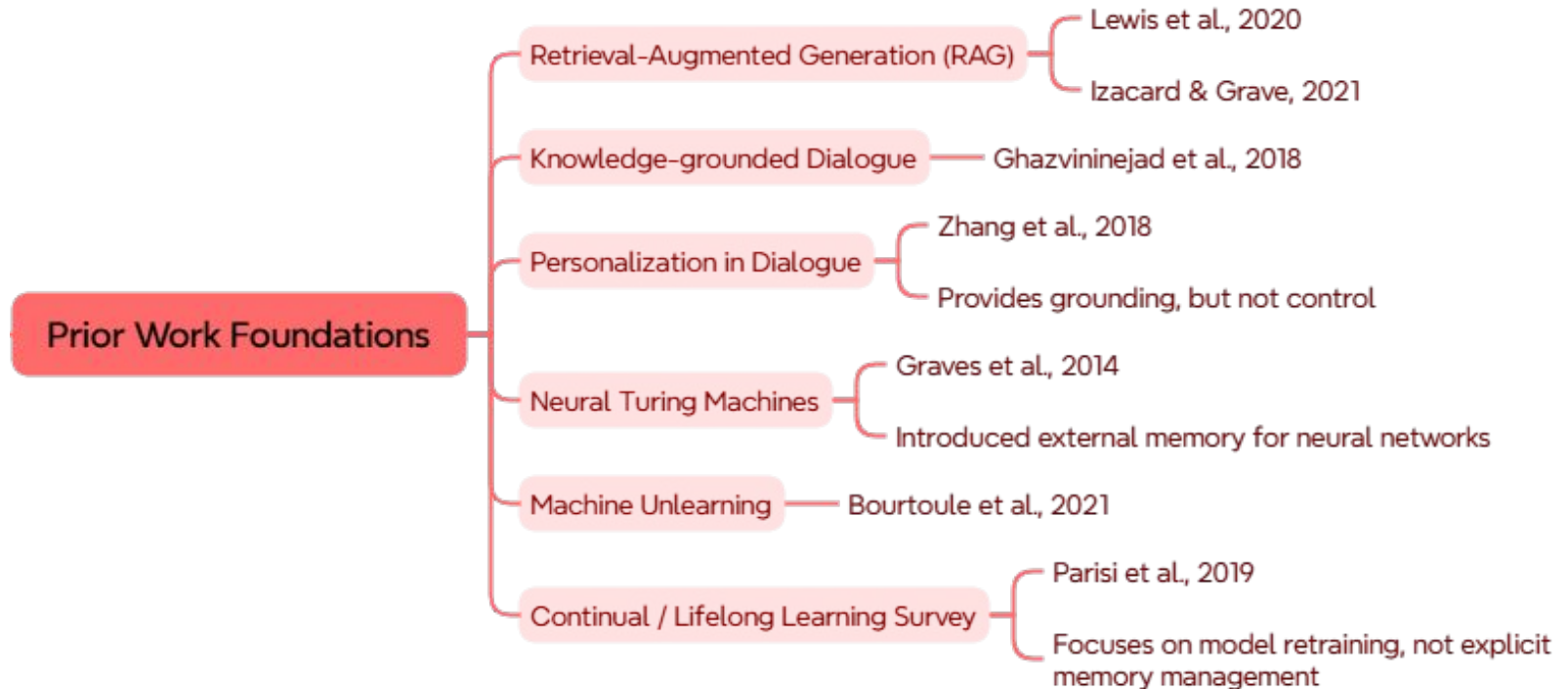
Controlled Evolution

Adapt to policy updates and resolve knowledge contradictions effectively.

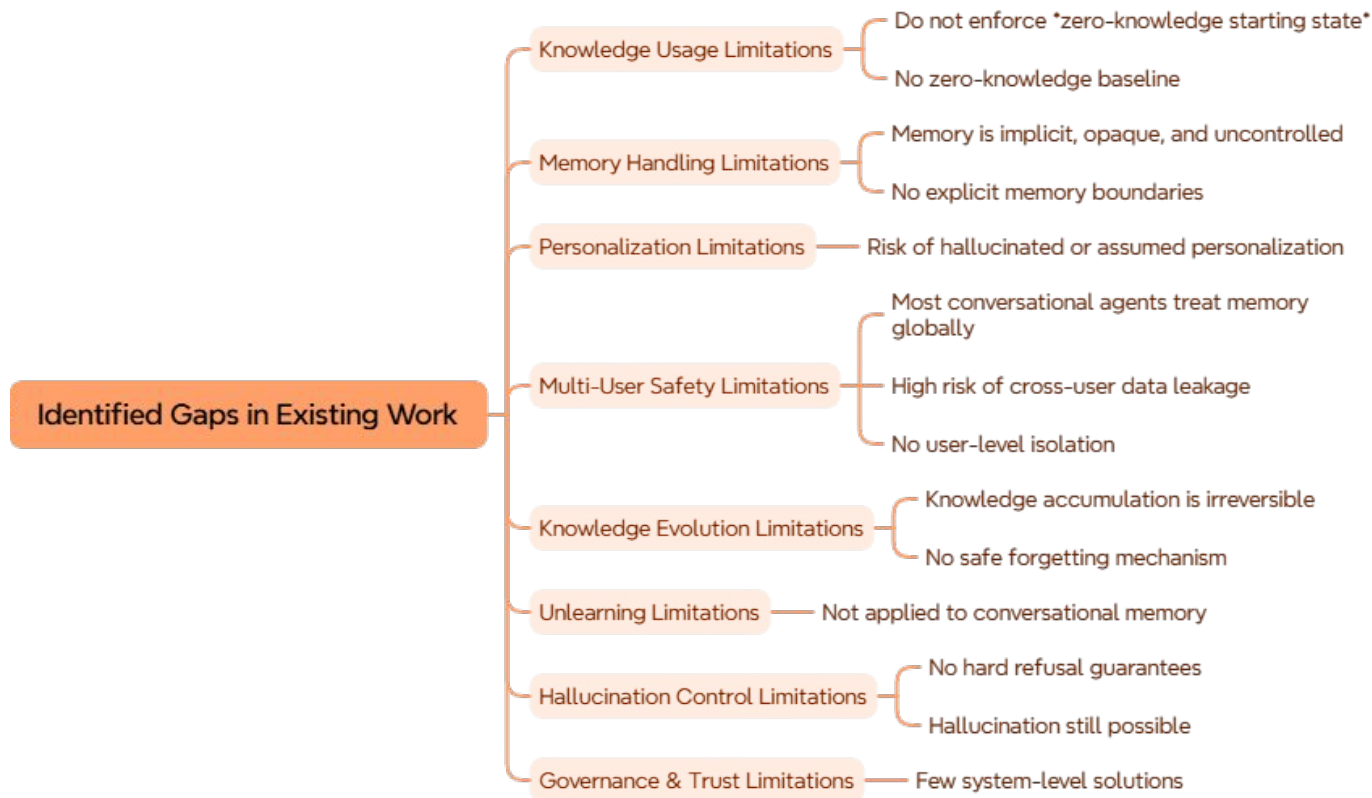
Research Context and Proposed AI Agent Architecture



Prior Work Foundations



Identified Gaps in Existing Work



Project's Novelties



Zero-Knowledge Operation

Answers strictly from verified knowledge with explicit refusal outside scope



Controlled Memory

Structured memory with validated creation, update, and deletion



Personalization Control

Personalization only from explicitly learned user context



Multi-User Isolation

Strict per-user memory separation with zero data leakage

Project's Novelties (contd.)



Knowledge Evolution

Confidence-driven updates and correction of stored knowledge



Unlearning Mechanism

Selective removal of incorrect knowledge without retraining



Hallucination Control

Deterministic refusal ensures only grounded responses



Governance & Trust

Auditable system with clear and enforceable knowledge boundaries

Literature Survey

Paper & Year	Problem Addressed	Key Contributions / What Was Done	Limitations / Gap	Relevance to Our Work / Improvement
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (Lewis et al., 2020)	Limitations of LLMs in accessing and manipulating world knowledge during generation	• Proposes RAG: combines pre-trained seq2seq model (parametric) with dense vector retrieval from external source (Wikipedia) to ground outputs • Evaluates two RAG variants conditioning retrieval differently during generation • Achieves SOTA on multiple open-domain QA tasks	• Focuses on general knowledge retrieval (Wikipedia) only • No personalization or user-specific memory • No multi-user memory isolation • No memory update validation or forgetting mechanism	• RAG is foundational to our memory grounding mechanism but does not support multi-user adaptive memory, contradiction detection, confidence scoring, or unlearning.

Literature Survey

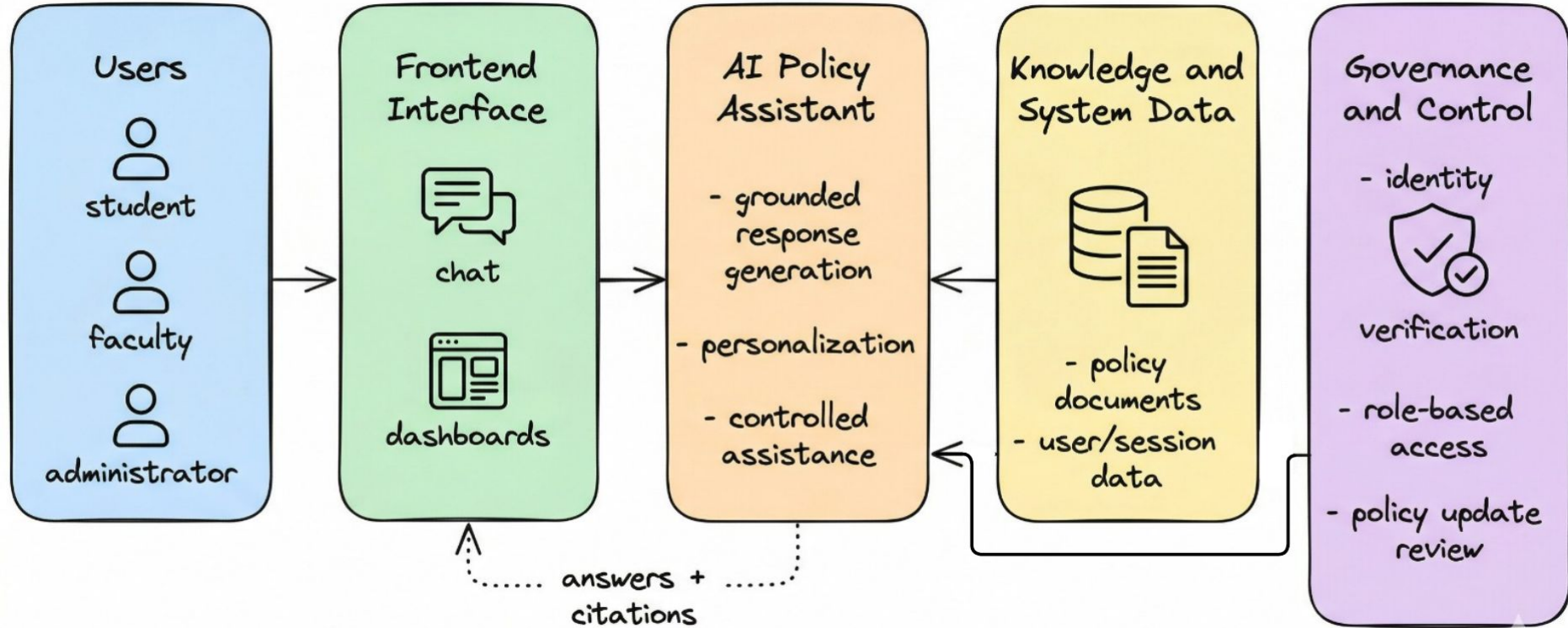
Paper & Year	Problem Addressed	Key Contributions / What Was Done	Limitations / Gap	Relevance to Our Work / Improvement
Continual Lifelong Learning with Neural Networks: A Review (Parisi et al., 2019)	The challenge of continuous/ongoing learning in neural networks without catastrophic forgetting	• Reviews continual learning approaches including memory replay, structural plasticity, etc. • Explains catastrophic forgetting as a central obstacle • Discusses cognitive inspirations for learning systems	• Survey of existing lifelong learning methods; no specific system implementation • Focus on classical neural networks, not modern LLMs • Doesn't address retrieval-augmented or vector memory	• Provides theoretical basis for why continuous adaptation and forgetting are difficult • Our work operationalizes these ideas with memory querying, selective updates, and unlearning mechanisms

Literature Survey

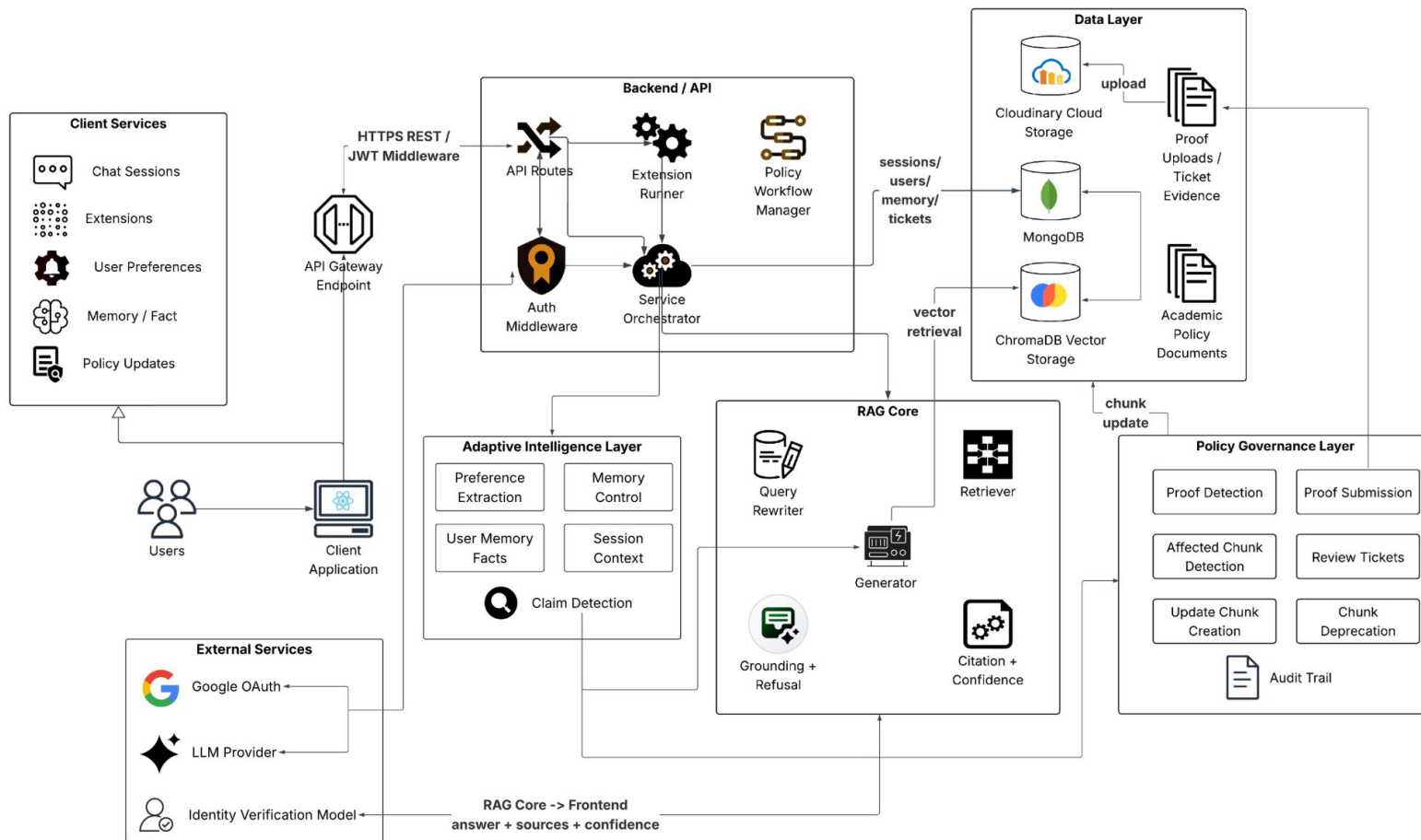
Paper & Year	Problem Addressed	Key Contributions / What Was Done	Limitations / Gap	Relevance to Our Work / Improvement
Leveraging Passage Retrieval with Generative Models ... (Izacard & Grave, 2021)	Improve QA by combining retrieval and generation	• Introduces Fusion-in-Decoder (FiD): a retriever that conditions generation on multiple passages for QA	• Static retrieval corpus • No user personalization or long-term memory	Supports <i>retrieval relevance</i> but lacks adaptive personalized memory
Neural Turing Machines (Graves et al., 2014)	Neural networks with external memory	• Introduces differentiable memory outside network parameters for end-to-end learning	• Not scalable/practical for LLMs or conversational agents • No privacy/multi-user concern	Inspires external memory concept, but replaced by vector DB for scalability

System Design & Architecture

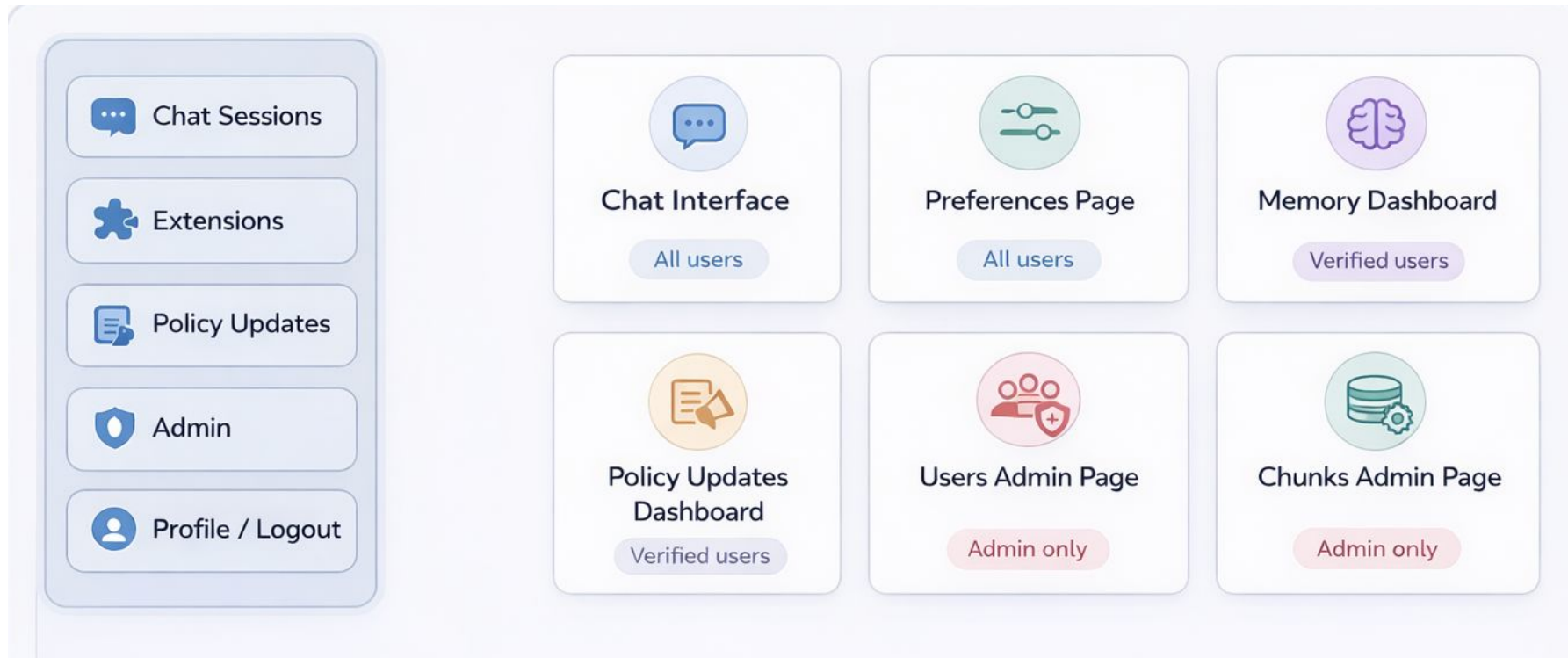
System Overview



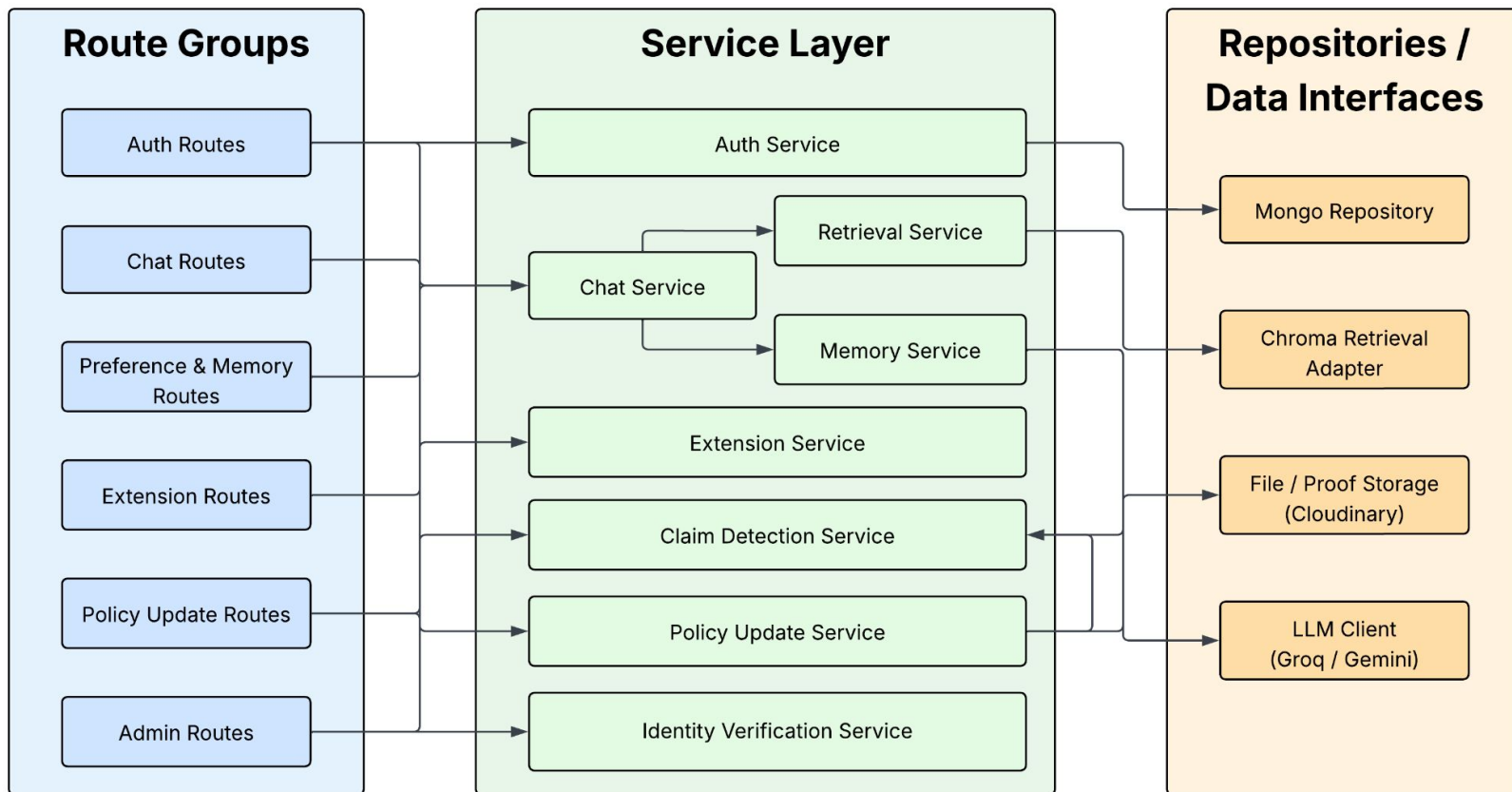
System Architecture



Frontend Overview

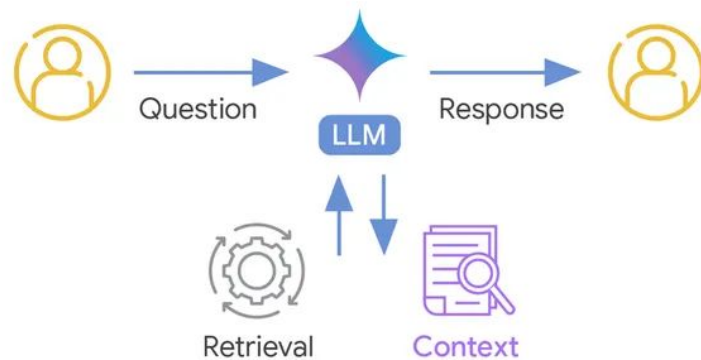


Finalised Backend and Service-Layer Architecture



Retrieval-Augmented Generation (RAG)

- RAG combines retrieval with generation
- Bridges the gap of parametric knowledge
- Grounded Response Generation
- Improves factual accuracy
- Supports domain-specific systems



Why RAG in our system?

Enforces Zero-Knowledge Baseline

Responds only from retrieved information, avoiding implicit model biases.

Reduces Hallucinations

Factual grounding minimizes unsupported or fabricated AI outputs.

Deterministic Refusal

System explicitly refuses to answer if relevant context is unavailable.

Scalable Knowledge Updates

Update the database instantly without retraining the core LLM.

Source-Grounded Responses

Enables traceability and verification by backing answers with context.

Strict Domain Control

Restricts knowledge to curated documents, preventing out-of-scope replies.

Memory Integration

Extends retrieval with user-specific memory for personalized context.

Auditable and Governed AI

Clear separation between model and knowledge ensures transparency.

Database (Simulated Policy Documents)



academic_advising_
and_mentorship.md



continuous_assessment_
_overview.md



course_add_drop_and_
withdrawal.md



exam_registration_
process.md



final_year_project_
guidelines.md



grading_components_a
nd_weightage.md



internship_registration_
and_evaluation.md



lab_evaluation_and_
internal_marks.md



project_submission_
and_review_flow.md



revaluation_and_answer_
script_review.md

Adaptive Intelligence Modules

Features in our System

User Authentication

Automatic Preference Extraction

Re-writing queries

Controlled Policy Update Workflow

Multi-user memory isolation

Context-aware fact extraction

Proactive Risk Detection

Extension-based academic tools

Canonical vs Adaptive RAG

Dimensions	Canonical RAG	Adaptive RAG
Core Objective	Accurate policy QA from fixed corpus	Accurate policy QA + personalized interaction
Knowledge source	Academic policy documents only	Same academic policy documents only
Personalization	Not available	Preferences + user fact memory applied at response time
Authentication	Minimal/basic flow	Multi-user auth with user-linked sessions and data
Conversation handling	Session chat history for UX	Context builder + query rewriting for follow-up clarity
Memory	No long-term user memory	Controlled memory (facts, retention, update/delete/lock flows)

Canonical vs Adaptive RAG (contd.)

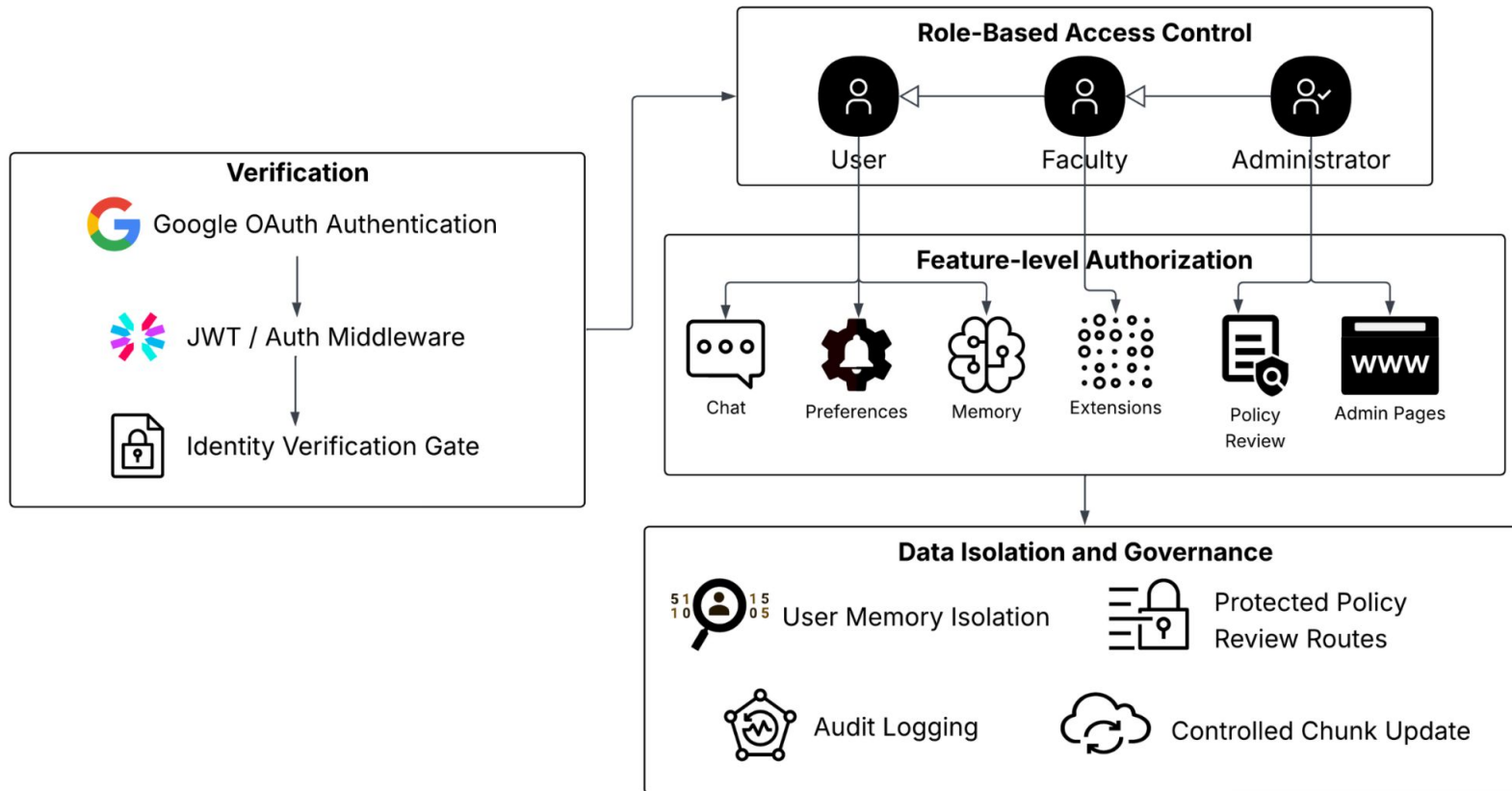
Dimensions	Canonical RAG	Adaptive RAG
Preference handling	None	Automatic extraction + application of response style preferences
Fact extraction	None	LLM-based fact extraction from user messages
Risk awareness	None	Conversation-level risk detection and alert metadata
Refusal behavior	Zero-knowledge refusal on insufficient context	Same refusal policy retained with adaptive wrapper
Hallucination control	Prompt-level constraints	Prompt constraints + validation checks
Key claim	Grounded answers	Grounded answers with adaptation, without changing corpus authority

Extensions

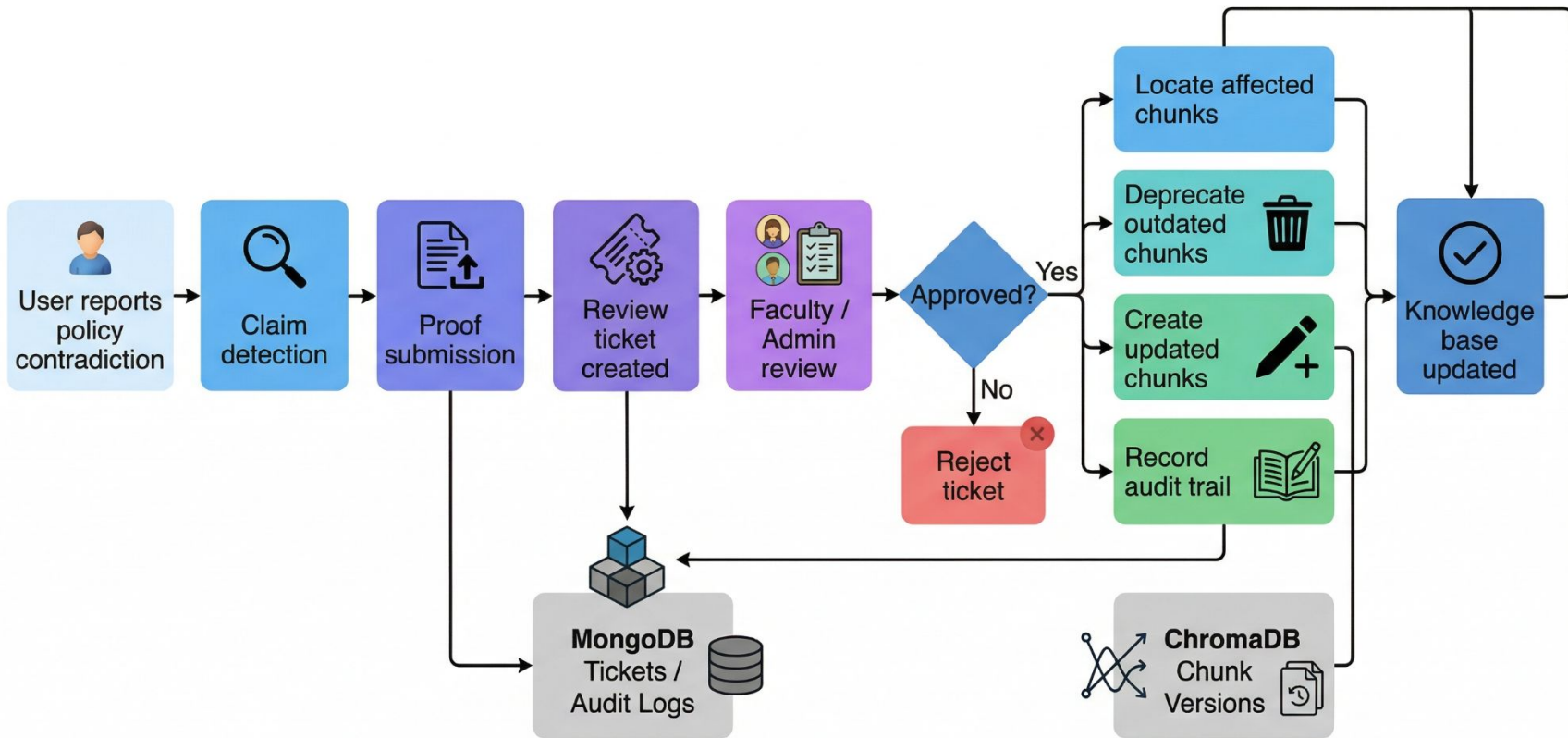
- The platform now supports specialized academic tasks through extensions in addition to normal policy-based question answering.

Aspect	Prompt-based Extension	Script-based Extension
Execution style	Works by injecting specialized prompts and instructions into the AI workflow	Works by executing predefined Python logic for a specific academic task
Best suited for	Good for conversational guidance, analysis, explanation, or role-specific assistance	Good for structured processing, file handling, calculations, and repeatable task automation
Control level	More flexible and natural-language driven, but dependent on model behavior	More deterministic and rule-driven because execution follows explicit program logic

Security, Roles, and Governance Controls



Controlled Policy Update Workflow



Policy Governance and Knowledge Evolution

'A governed pipeline for policy evolution within the academic assistant...'

- Claim Detection
- Contradiction Analysis
- Proof Submission
- Ticket Creation
- Affected Chunk Identification
- Chunk Deprecation
- Updated Chunk Creation
- Audit Logging

System Performance Metrics

Dataset Validation

93.1%

Success Rate

98.7%

Grounded Response
Fidelity

0.6784

Refusal Correctness

1.0

Hallucination Rate

0.0463

Personalization
Accuracy

0.7391

Zero-Knowledge
Compliance

1.0

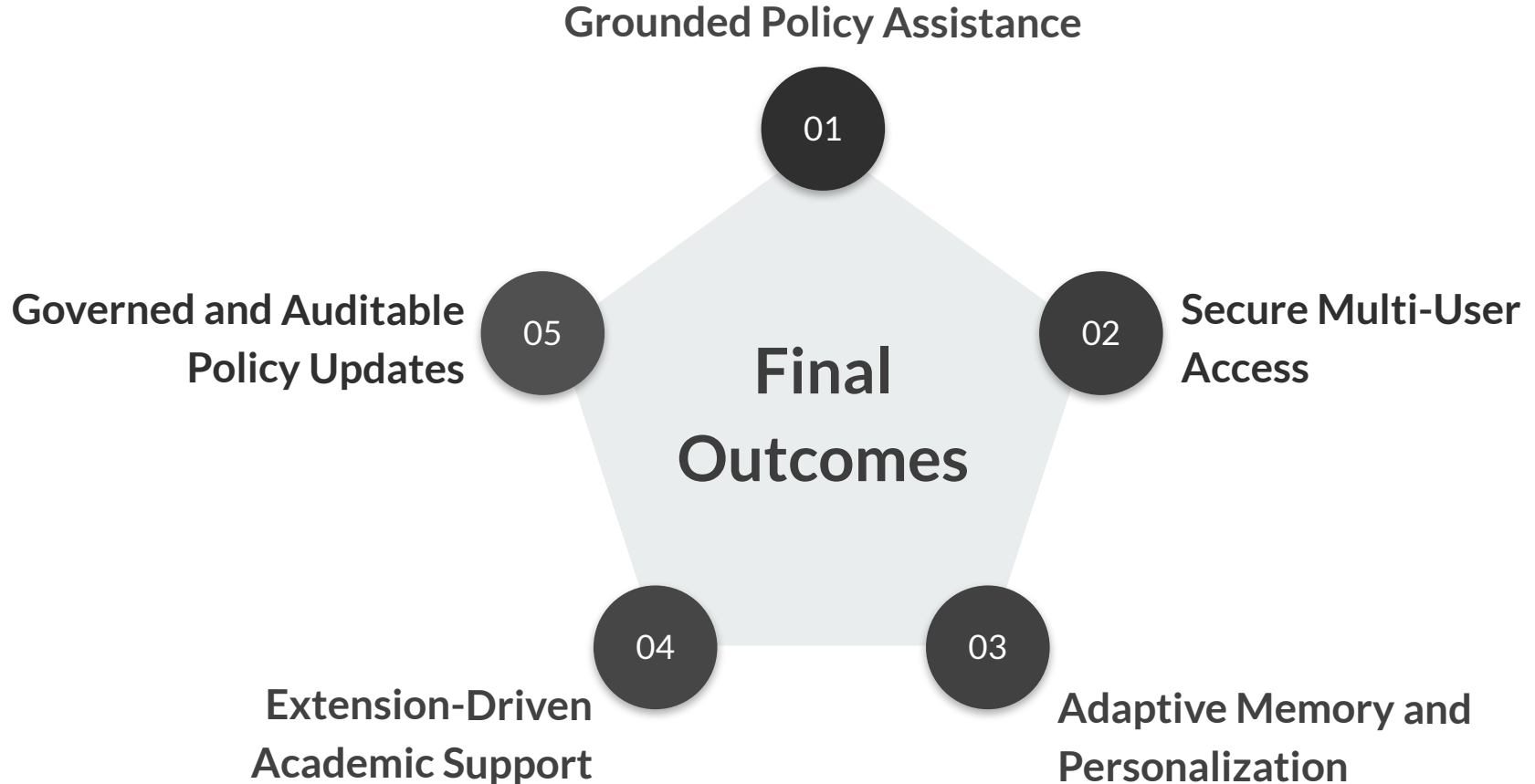
Contradiction
Resolution Rate

1.0

Dataset Validation done by Claude Sonnet 4.5

Application Validation done by OpenAI gpt-4.1-mini

Final Outcomes and Significance



Thank You!